



January 22, 2026

Ms. Lindsey Wells
Michigan Department of Environment, Great Lakes, and Energy
Air Quality Division, Cadillac District Office
120 West Chapin Street
Cadillac, MI 49601

Subject: Test Plan for a Compliance Performance Demonstration of One (1) Utility
(Open) Flare at Wexford County Landfill, Manton, Michigan (ROP MI-ROP-
N3862-2022)

Dear Ms. Wells:

On behalf of Wexford County Landfill (Wexford), Environmental Information Logistics, LLC. (EIL), is pleased to provide this test plan for a compliance performance demonstration of one (1) utility flare at Wexford County Landfill in Manton, Michigan. The purpose of this document is to notify the Michigan Department of Environment, Great Lakes, and Energy (EGLE) of the proposed test date and provide the required source and test program information, in accordance with R 336.1213(3), 40 CFR 63.11(b) and 40 CFR 60.18(f).

EIL has been contracted by Wexford to perform this project. Mr. Benjamin Kotrba, Environmental Scientist with EIL, prepared this test plan.

The field test is tentatively scheduled for Friday, February 27, 2026. This test plan is formatted in accordance with EGLE "*Format for Submittal of Source Emission Test Plans and Reports (November 2019)*":

1a. Names, titles, and telephone numbers for personnel involved with this study are:

Name and Title	Company	Telephone
Benjamin Kotrba Environmental Scientist	Environmental Information Logistics, LLC. 130 E. Main Street Caledonia, Michigan 49316	(989) 415-3741
Chris Gee General Manager	GFL Environmental 1307 Higgins Drive Cheboygan, Michigan 49721	(231) 597-8553

Wexford County Landfill is located at 990 US Highway 131 North, Manton, Michigan 49724.

- 1b. The expected test date at Wexford will be performed on Friday, February 27, 2026.
- 1c. Wexford is a municipal solid waste landfill. Wexford employs a utility (open) flare to control landfill gas emissions.
- 1d. Wexford is not a production operation: there are no finished products produced. The only raw material involved is landfill gas.
- 1e. The operating parameters used to regulate the utility flare are landfill gas flow rate and proof of flame.
- 1f. The utility flare has a rated capacity of 1,600 standard cubic feet per minute (scfm) of landfill gas.
- 2a. The utility flare currently serves as the primary control device for landfill gas.
- 2b. The utility flare operates at a varied landfill gas flow rate depending on flow rate from the waste mass. Landfill gas flow rate to the flare is delivered by a blower.
- 2c. The utility flare is designed to meet the requirements of 40 CFR 60.762(b)(2)(iii) and 63.1959(b)(2)(iii) at a flow rate of up to 1,600 scfm. The utility flare is expected to be tested at approximately 900 - 1,300 scfm, which are typical operating conditions for the flare when in operation.
- 2d. There has been no significant maintenance activity within the last three months.
- 3. The utility flare must show compliance with the provisions of §60.18 and §63.11.
- 4. EIL will measure landfill gas flow rate, gas composition and moisture content (for net heating value analysis), and visible emissions from the utility flare.
- 5. EIL will measure landfill gas flow rate to the utility flare by USEPA Methods 1A and 2C. EIL will analyze the landfill gas composition by USEPA Method 3C. EIL will extract integrated canister samples and have them analyzed for carbon dioxide, methane, nitrogen, and oxygen. EIL will calculate the landfill gas net heating value from Method 3C analysis results. EIL will measure the landfill gas moisture content by USEPA Method 3C (included in analysis).

Utility flare sample collection and analysis methods include:

Parameter	Method	Analytical Method
Landfill Gas Flow Rate; Flare Exit Velocity	USEPA Methods 1A& 2C	Field Data
Gas Composition and Moisture Content; Net Heating Value	USEPA Method 3C	Gas Chromatography / Thermal Conductivity Detector (GC/TCD)
Visible Emissions	USEPA Method 22	Observation

EIL will use Method 1A, “*Sample and Velocity Traverses for Stationary Sources with Small Ducts or Stacks*,” to determine the appropriate number and location of traverse points at each utility flare inlet. These determinations will be based on dividing the stack cross-section into equal areas (one traverse point for each area) and the number of upstream and downstream stack diameters from the sample ports to the nearest flow disturbance.

EIL will use Method 2C, “*Determination of Stack Gas Velocity and Volumetric Flow Rate in Small Stacks or Ducts (Standard Pitot Tube)*,” to measure velocity pressures and temperatures at each traverse point. EIL will use a standard pitot tube, with a baseline coefficient of 0.99, positioned at each traverse point. The velocity pressure and exhaust gas temperature will be measured and recorded. Due to extremely high static pressure, EIL intends to use a digital manometer for velocity pressure measurements. The digital manometer will have increments of 0.01 inch of water. Temperature measurements will be made using a calibrated temperature probe.

The average stack gas velocity is a function of the average velocity pressure, absolute stack gas pressure, average stack gas temperature, molecular weight of the wet stack gas, and pitot tube coefficient. The derivation of the average stack gas velocity will be calculated using the equations stipulated in this test method. The actual stack gas flow rate will be calculated using the average stack gas velocity and the cross-sectional area of the stack.

EIL will use Method 3C, “*Determination of Carbon Dioxide, Methane, Nitrogen, and Oxygen from Stationary Sources*,” to determine the net heating value of the landfill gas. EIL will collect a minimum of three, 30-minute, integrated samples of landfill gas from the utility flare inlet. EIL will contract Enthalpy Analytical, Richmond, Virginia, to analyze each sample for carbon dioxide, (CO₂), methane (CH₄), nitrogen (N₂), and oxygen (O₂) concentration and moisture fraction. Figure 1 details the Method 3C sample train.

EIL will use the Method 3C analysis data to calculate stack gas molecular weight for stack gas velocity and volumetric flow rate calculations, and to calculate the net heating value of the gas being combusted per §63.11(b)(6)(ii). The net heating value will be the arithmetic average of the three test runs.

Enthalpy will follow the analytical procedures of Method 3C by using a gas chromatograph (GC), with appropriate separation column for the expected parameters, equipped with a thermal conductivity detector (TCD).

The dry molecular weight of the stack gas will be calculated based on the assumption that the primary constituents are methane, carbon dioxide, nitrogen, and oxygen (other compounds present have a negligible relative concentration). The stack gas dry molecular weight is equal to the sum of stack gas constituent concentrations (%) multiplied by the corresponding molecular weight of that constituent. As noted above, the moisture fraction will be determined by the Method 3C analysis.

EIL will use Method 22, *“Visual Determination of Fugitive Emissions from Material Sources and Smoke Emissions from Flares,”* to determine the presence and duration of visible emissions. EIL will conduct one, 120-minute, non-continuous observation of the utility flare exhaust for smoke emissions. EIL will observe continuously for 15 to 20 minutes, break for at least 5 - but no more than 10 minutes, and resume observation until a full 120-minutes of observation time has accrued. Data will be recorded on the data sheet shown as Figure 22-1 of Method 22.

6. The utility flare test program will be comprised of triplicate 30-minute sample runs for net heating value, and a single, 120-minute observation of visible emissions. EIL will conduct velocity traverses during the scheduled visible observation breaks.
7. Figures 2 and 3 depict a representative utility flare skid, and inlet traverse point locations, respectively.
8. The estimated gas composition of the landfill gas to the utility flare is 45 to 55% methane, 40 to 55% carbon dioxide, and less than 5% combined nitrogen, oxygen, and trace volatiles. This will yield a net heating value of at least 14.97 megajoules per standard cubic meter (MJ/scm) which is greater than 7.45 MJ/scm for non-assisted flares.
9. The utility flare will operate at the maximum routine operating condition, based on site conditions (landfill gas production and quality). Wexford anticipates that the utility flare will operate at approximately 900 - 1,300 scfm.
10. Landfill gas flow rate and proof of flame will be monitored and recorded at least once every 15 minutes on the utility flare.
11. N/A

12. EIL will prepare and calibrate field measurement equipment and perform quality assurance/quality control (QA/QC) consistent with USEPA methodology.
13. EIL will sample and analyze audit gas cylinders for methane, carbon dioxide, nitrogen, and oxygen, if provided by EGLE. Audit gases should be within the expected concentration ranges outlined in Section 6 of this test plan. EIL will not assume any role or responsibility in transport of any audit cylinder to, or from, the test site.

The emission test report will include the items found on pages 3 and 4 of the EGLE Air Quality Division's *Format for Submittal of Source Emission Test Plans and Reports*. Included in the report will be a site description with the reason for tests, a source description, a summary of results, EIL sample collection and analytical procedures, test results, operational data, discussion of results, tables, figures, raw data sheets, Chain-of-Custody forms, the analytical laboratory report, and example calculations.

Please call me at (989) 415-3741 if you have any questions or require additional information.

Sincerely,

Environmental Information Logistics LLC

Benjamin A. Kotrba

Benjamin A. Kotrba
Environmental Scientist

Attachments

cc: Lindsey Wells – MI EGLE
Chris Gee – GFL Environmental
Wexford County Landfill Operating Records

FIGURES

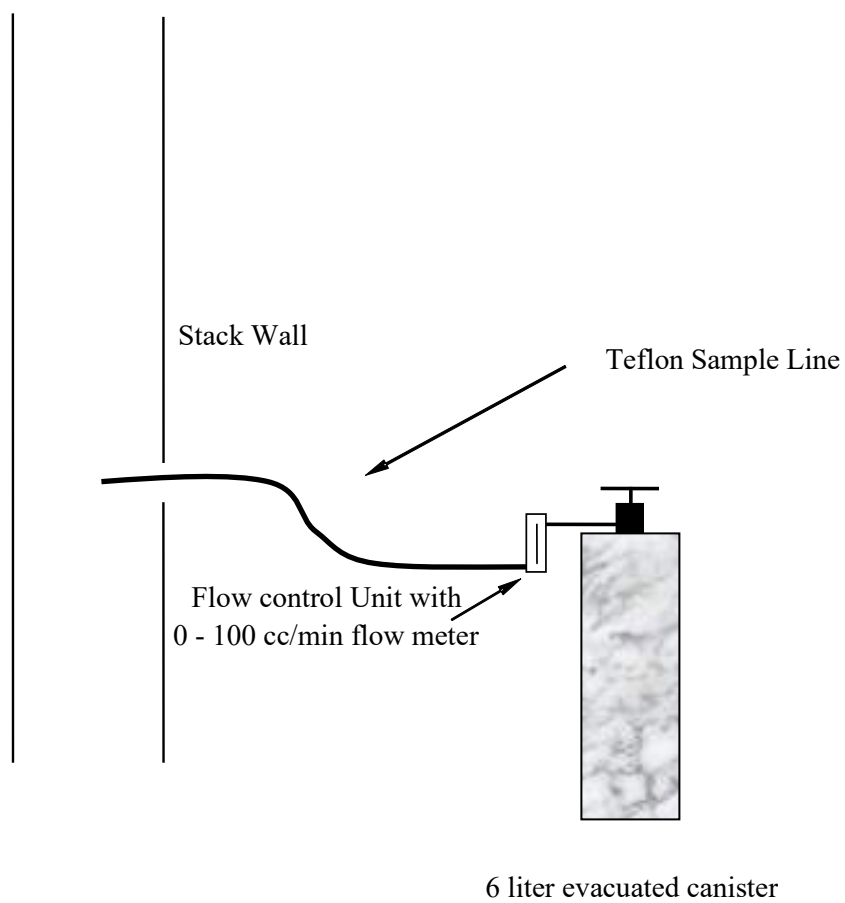


Figure 1

USEPA Method 3C sample train for the utility flare inlet duct at
Wexford County Landfill in Manton, Michigan.

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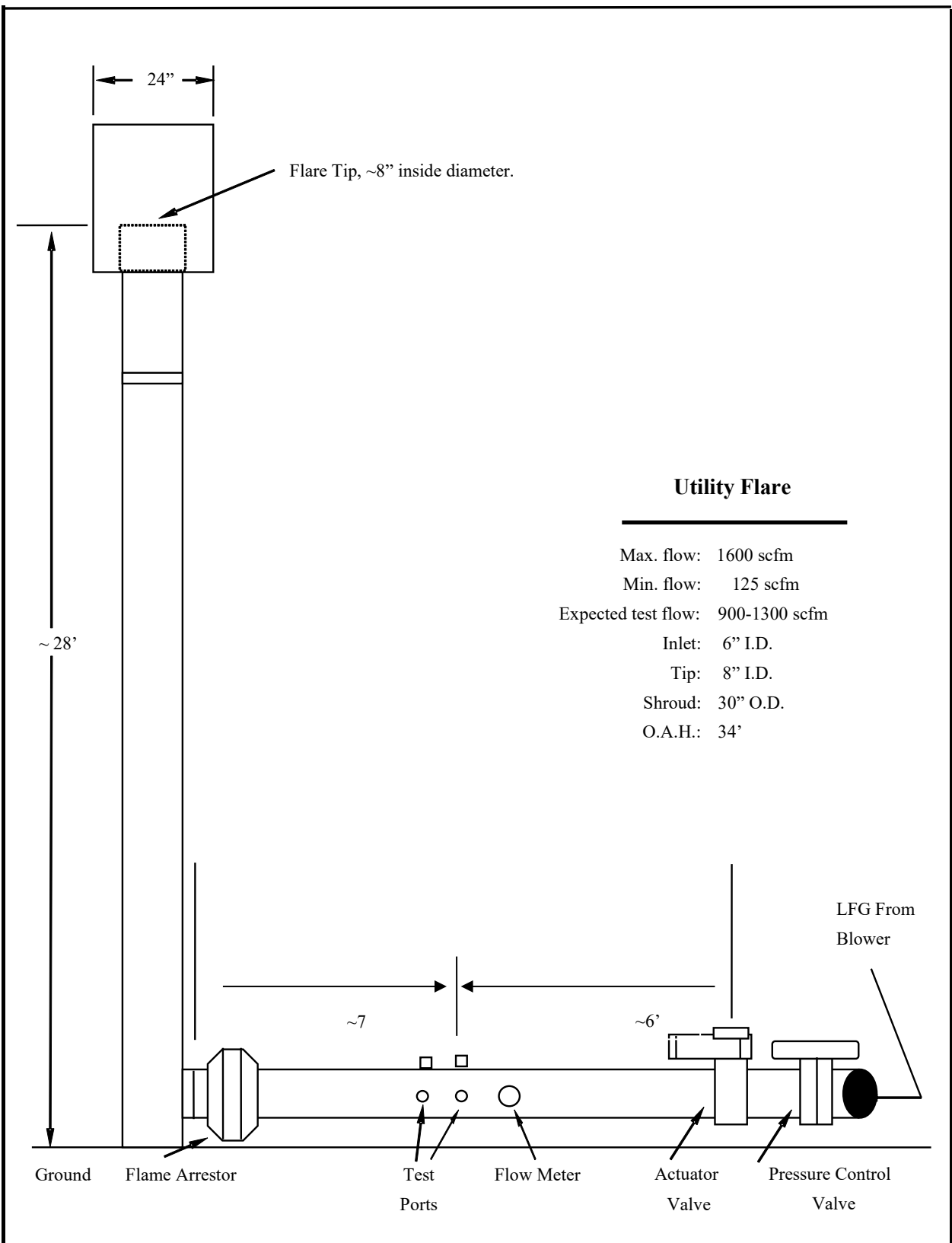
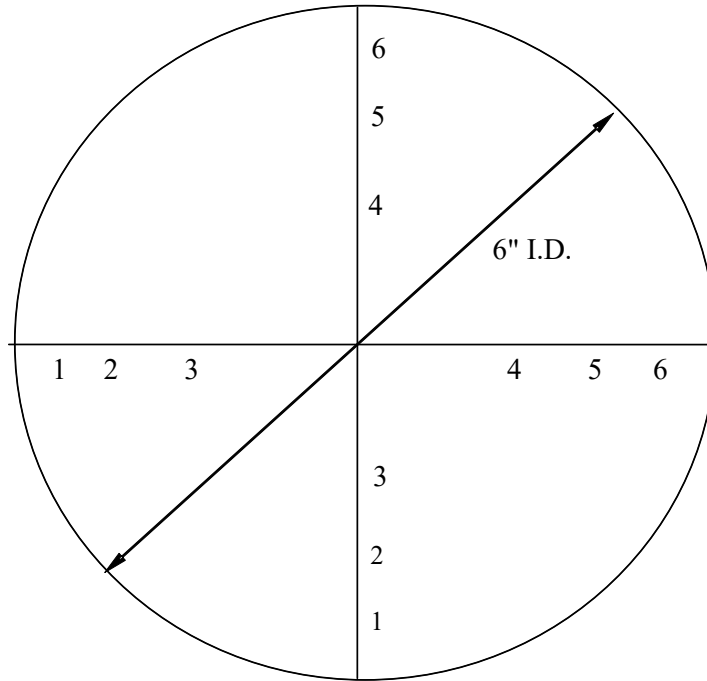


Figure 2

Utility flare duct and stack arrangement, approximate dimensions, and test locations, Wexford County Landfill in Manton, Michigan.

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Traverse Point Number	Distance From Wall (inches)
1	0.50
2	0.88
3	1.78
4	4.22
5	5.12
6	7.50

Figure 3

Traverse point numbers and locations on the utility flare inlets
at Wexford County Landfill in Manton, Michigan.

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